ASSIGNMENT-1

Long Term Virtual Internship – Data Analytics with Tableau

- **Purpose:** Compare total sales (SUM of Total) across different Product Lines. Identify which product line has higher or lower sales Help in business decision-making, like focusing on top-selling products.

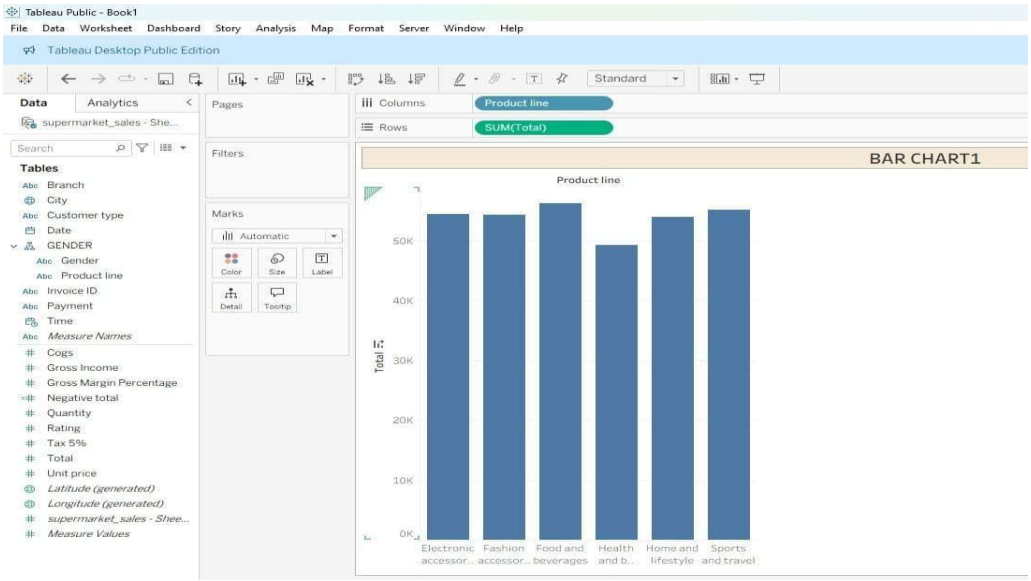
1.Prepare Bar Chart

- Step 1: Open Tableau Open Tableau Public / Tableau Desktop Connect to the dataset super market sales
- Step 2: Open a New Worksheet Click on Worksheet 1
- Step 3: Drag Dimensions and Measures Drag Product line → to Columns Drag Total → to Rows Tableau automatically applies SUM(Total)
- Step 4: Choose Bar Chart in the Marks card: Select Bar (or keep Automatic, Tableau selects Bar chart)
- Step 5: Format the Chart (Optional) Add Chart Title → BAR CHART1 Adjust colors if needed Resize bars for better visibility
- Step 6: Analyze the Chart Each bar represents a Product Line Height of the bar shows Total

Sales You can clearly compare sales across categories

Output:

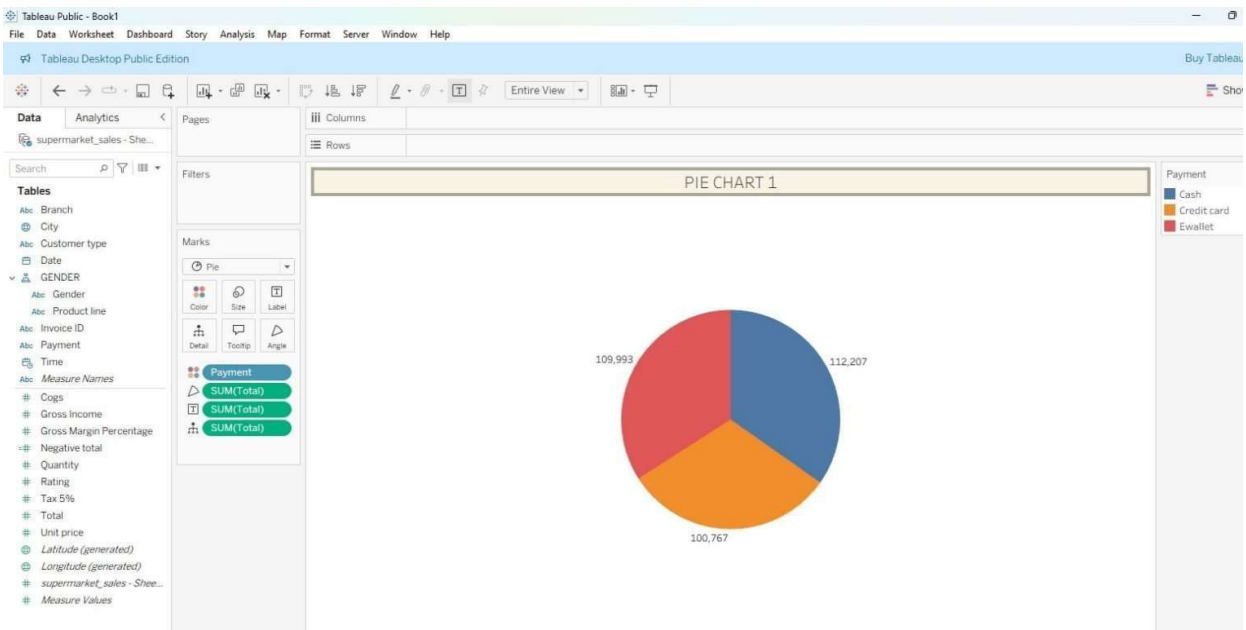
Bar chart:



2.Prepare pie chart:

- Step 1: Open Tableau Open Tableau Public / Tableau Desktop Connect to the dataset super market sales
- Step 2: Open a New Worksheet Click on Worksheet
- Step 3: Change Marks Type In the Marks card, change Automatic to Pie
- Step 4: Drag Fields to Marks Drag Payment → to Color Drag Total → to Angle Drag Total
  → to Label Drag Total → to Size Make sure aggregation is SUM(Total)
- Step 5: Format the Chart (Optional) Increase pie size using Size Show data labels (values or percentages) Add chart title → PIE CHART 1 Adjust colors for better clarity
- Step 6: Analyze the Chart Each slice represents a payment method Slice size shows total sales amount larger slice = more usage of that payment method

Pie chart:

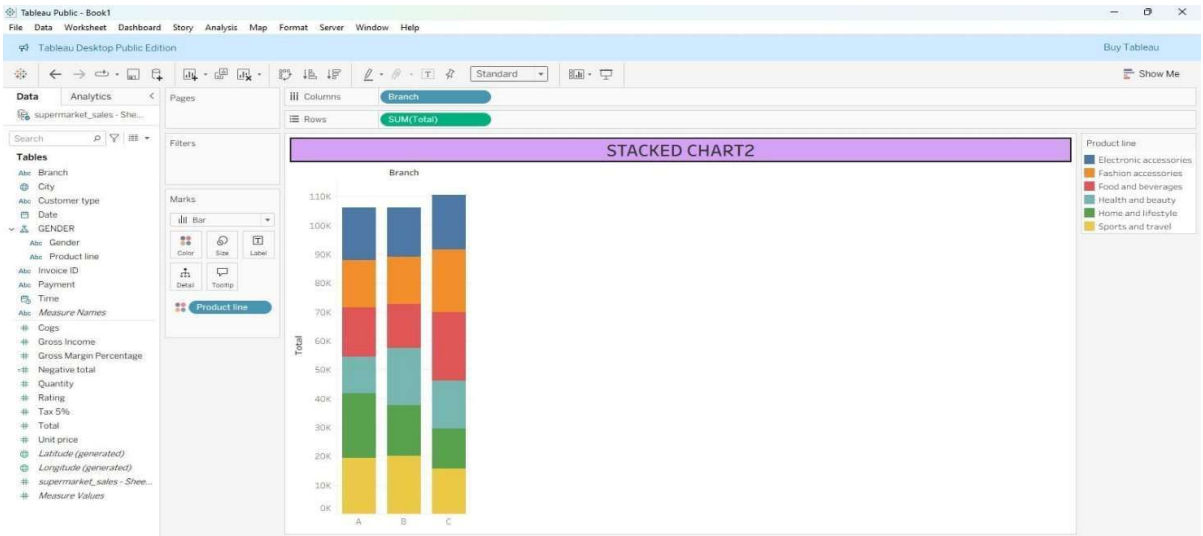


3.Prepare Stacked Bar Chart:

- Step 1: Open Tableau Open Tableau Public / Tableau Desktop Connect to the dataset super market sales
- Step 2: Open a New Worksheet Click on a new Worksheet
- Step 3: Drag Fields to Shelves Drag Branch → to Columns Drag Total → to Rows Tableau applies SUM(Total) automatically
- Step 4: Create Stacked Bars in the Marks card: Select Bar Drag Product line → to Color
- Step 5: Enable Stacking (if needed) Go to Analysis → Stack Marks → On
- Step 6: Format the Chart Add title → STACKED CHART2 Adjust colors for each product line Resize bars for clarity

- Step 7: Analyze the Chart Each bar represents a Branch (A, B, C) Each color segment shows a Product Line Total height of bar = Total Sales of that branch.

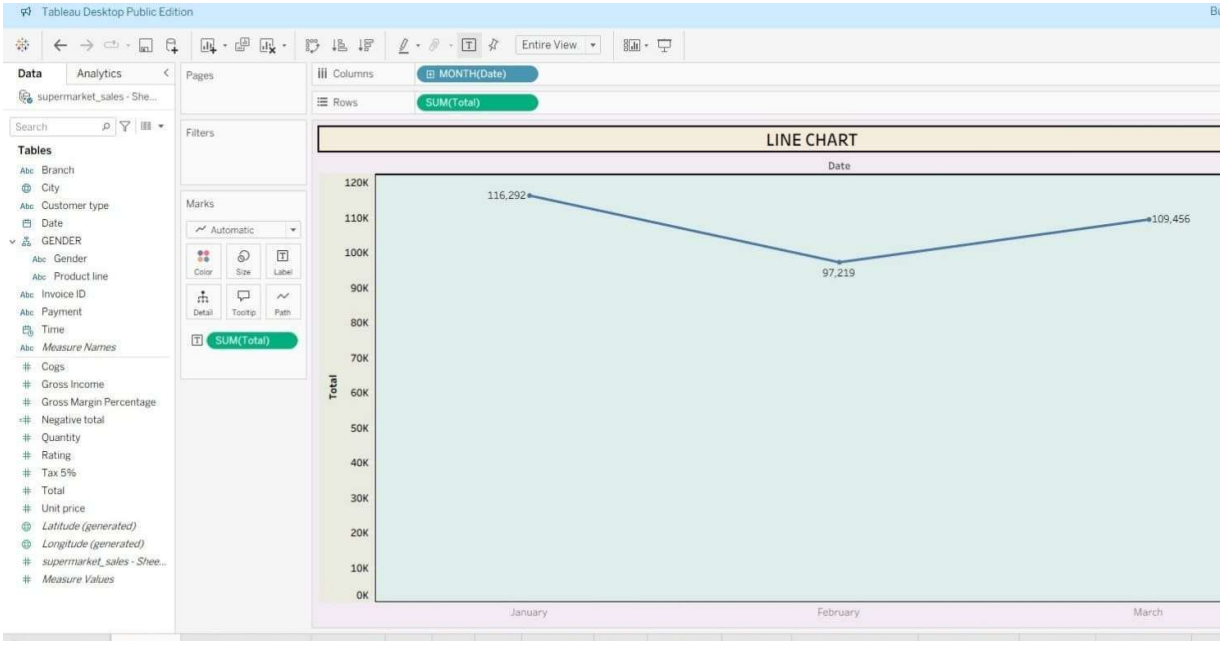
Stacked Bar Chart:



4.Prepare Line Chart:

- Step1: Open Tableau and drag the “super market sales” data set onto the canvas.
- Step2: Drag the “Date” field onto the Columns shelf and convert it to MONTH(Date) using the blue pill option.
- Step3: Drag the “Total” field onto the Rows shelf and apply the SUM aggregation.
- Step4: In the Show Me panel select the line chart (or let Tableau automatically assign it).
- Step5: Set the Marks card to “Automatic” and optionally add Label to display exact values on the line.
- Step6: Drag relevant dimensions (e.g., Branch, City) to the Filters shelf if you want to narrow the analysis.
- Step7: Enable Tooltip or adjust formatting (colors, size) for better readability.

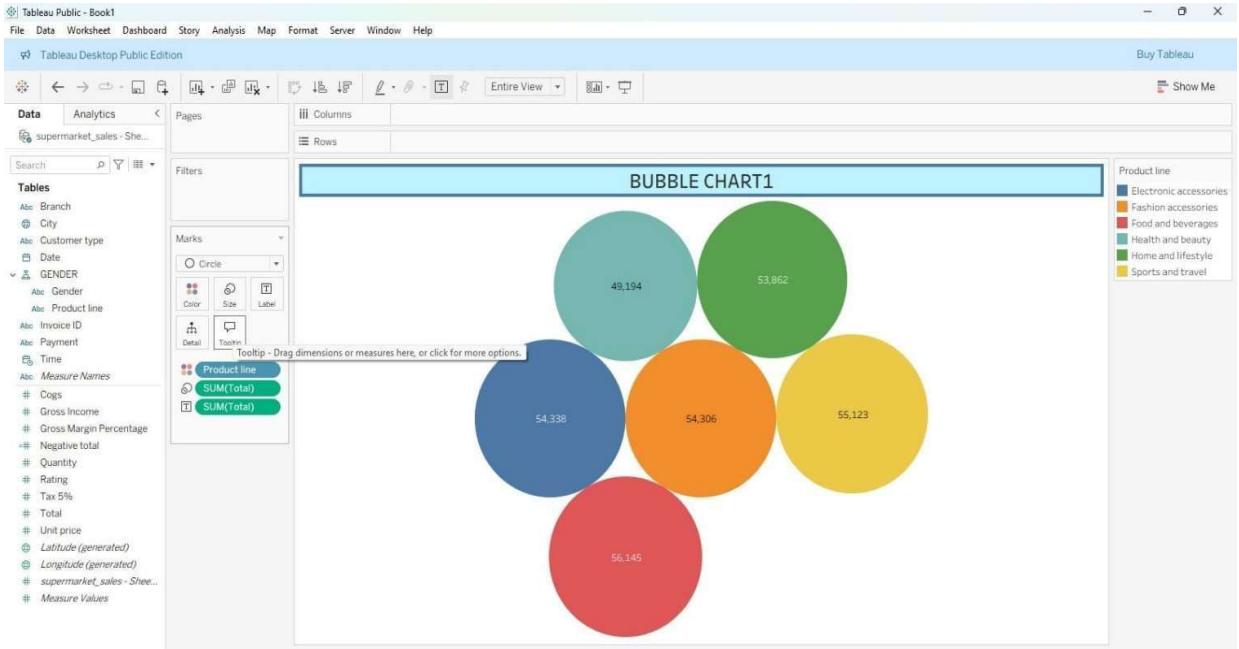
Line Chart:



Prepare Bubble chart:

- Step1: Open Tableau and connect to your dataset (here, “supermarket
- Step2: Drag “Product line” to *Color shelf*. Assigns each product line a unique color for differentiation.
- Step3: Drag “SUM(Total)” to *Size shelf* determines the size of each bubble based on total sales.
- Step4: Drag “SUM(Total)” again to *Label shelf*displays the total sales value inside each bubble sales”).
- Step5: Change mark type to *Circle (bubble)* Converts the visualization into a bubble chart.
- Step6: Place “Product line” on *Detail (optional)* Ensures each bubble represents a distinct product line.
- Step7: Adjust layout & title: Add a title (“BUBBLE CHART1”) and format as needed.

Bubble chart:



Assignment-2

Long Term Virtual Internship Data Analytics using Tableau

1.Prepare Donut Chart:

Step 1: Create a New Sheet and Rename as “Donut chart

Step 2: Open a New Worksheet Click on Worksheet 1

Step 3: Click Marks card -Select Pie from dropdown-Drag

Product line → Colour

Total → Angle

Total→ Label

Step 4: Create → Calculated Field Name it Donut Enter formula:0click ok

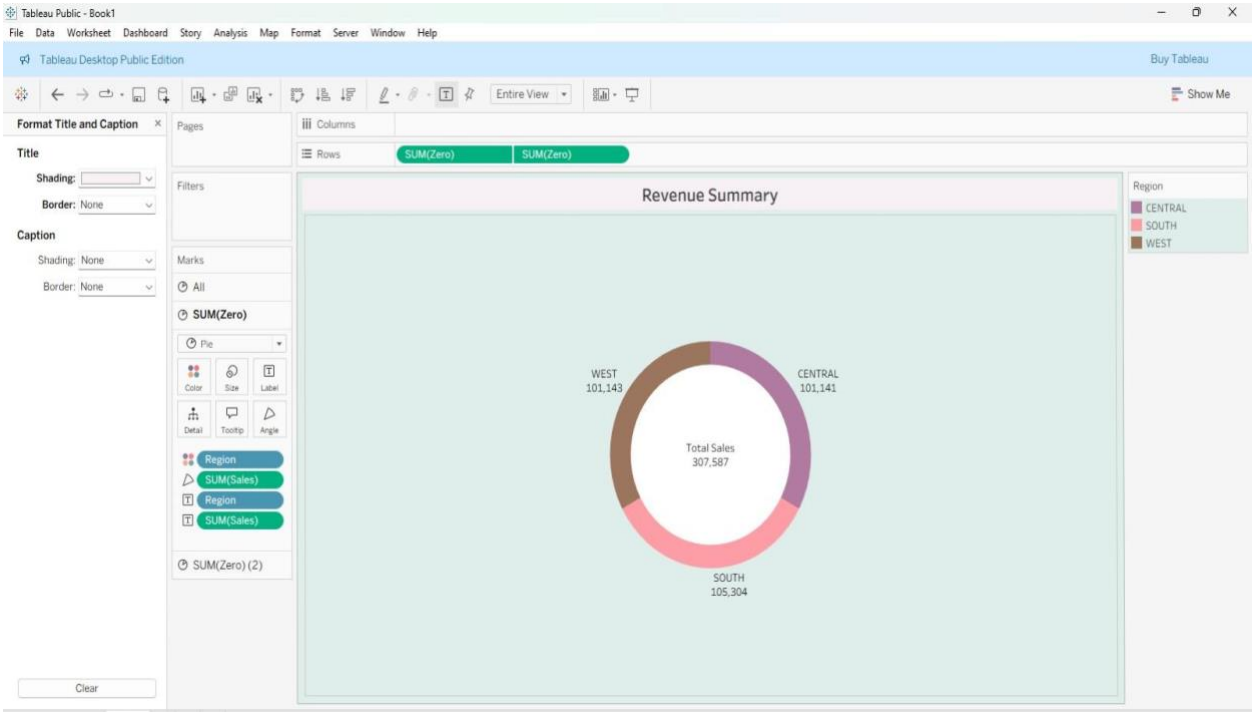
Step 5: Drag Donut → Rows (twice) (So Donut appears two times in Rows)

Right-click on second Donut → Select Dual Axis

Right-click again → Synchronize Axis

Step 7: Drag payment to colours

Donut chart: revenue summary



2.prepare Area chart

Step1: create a new sheet renamed as Area chart

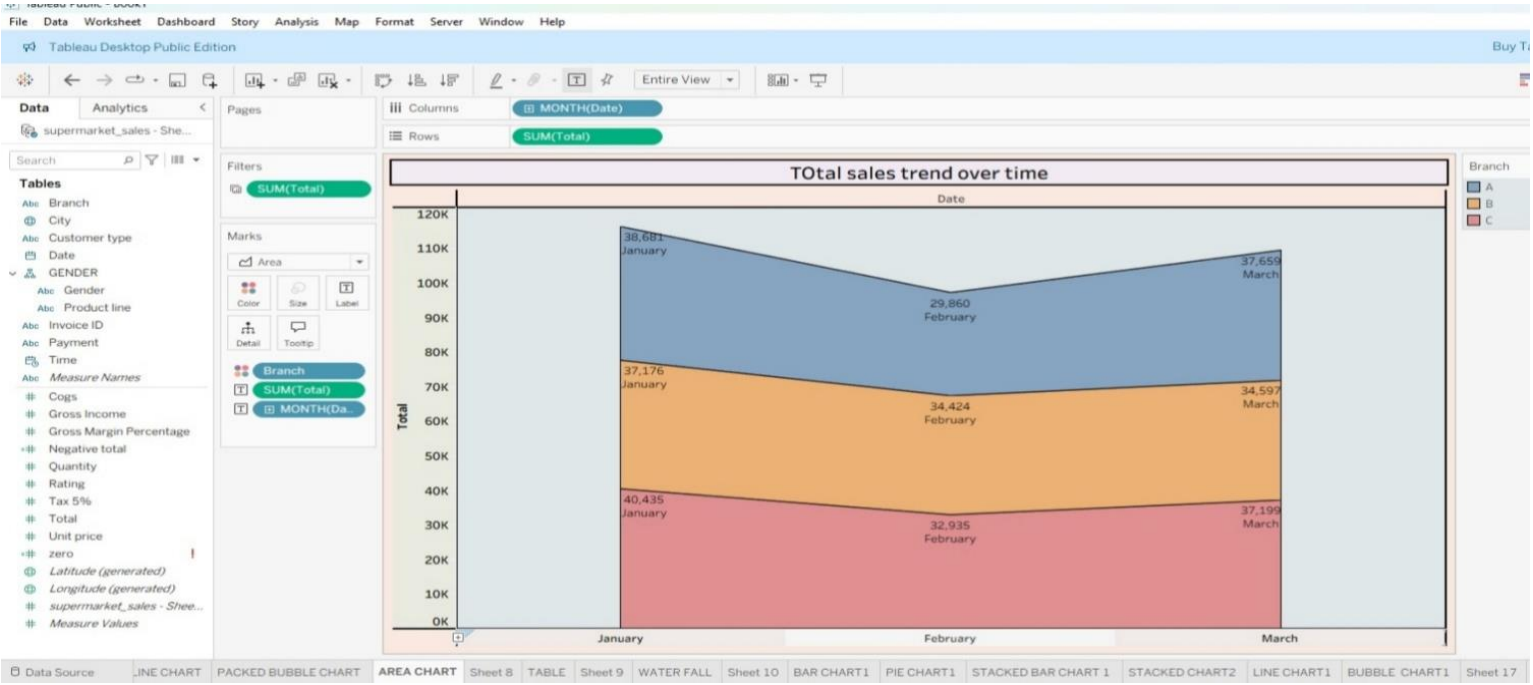
Step 2: Drag Date → Columns shelf -Click the drop-down on Date- Select Month

Step-03: Drag Total → Rows shelf

Step-04: Click Show Me (top right) Select Area Chart

Step-05: Drag city → Colour (in Marks card)

Area Chart: Total sales trend over time



3.prepare text table

Step1: Create a New Sheet and Renamed as “Text table”.

Step-02: On the right side, find Show Me. Click the Text Table

Step–03: Drag data column

Date → Columns shelf

Step – 04: From the Data Pane (left side), drag these fields to Rows

•Invoice ID

- Branch
- Customer type

•Gender

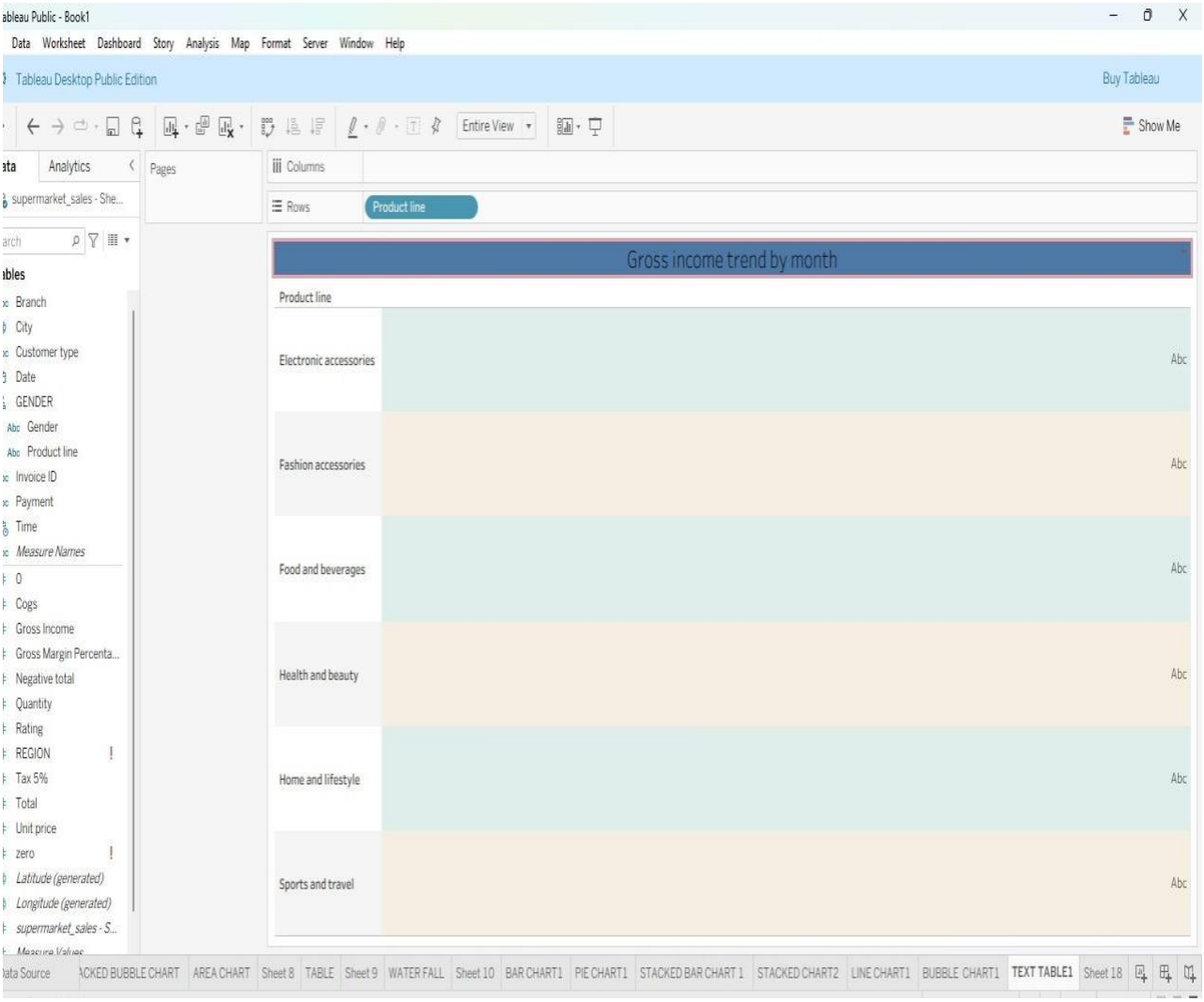
- Product line
- Date

Step 5 Drag Measures to Columns (or Text)

Step–06: Now drag these Measures: To Columns:

- Unit price
- Quality
- Tax
- Total
- Cogs

Text table: Gross income trend by month


ableau Public - Book1
Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Tableau Desktop Public Edition
Buy Tableau
Entire View
Show Me
Analytics
Pages
Columns
Rows
Product line
supermarket_sales - She...
Branch
City
Customer type
Date
GENDER
Gender
Product line
Invoice ID
Payment
Time
Measure Names
0
Cogs
Gross Income
Gross Margin Percenta...
Negative total
Quantity
Rating
REGION
Tax 5%
Total
Unit price
zero
Latitude (generated)
Longitude (generated)
supermarket_sales - S...
Gross income trend by month
Product line
Electronic accessories
Fashion accessories
Food and beverages
Health and beauty
Home and lifestyle
Sports and travel
Abc
AREA CHART
Sheet 8
TABLE
Sheet 9
WATER FALL
Sheet 10
BAR CHART1
PIE CHART1
STACKED BAR CHART 1
STACKED CHART2
LINE CHART1
BUBBLE CHART1
TEXT TABLE1
Sheet 18

4.PrepareHighlightedTable

Step 1: Create a New Sheet and Renamed as “Highlighted Table”

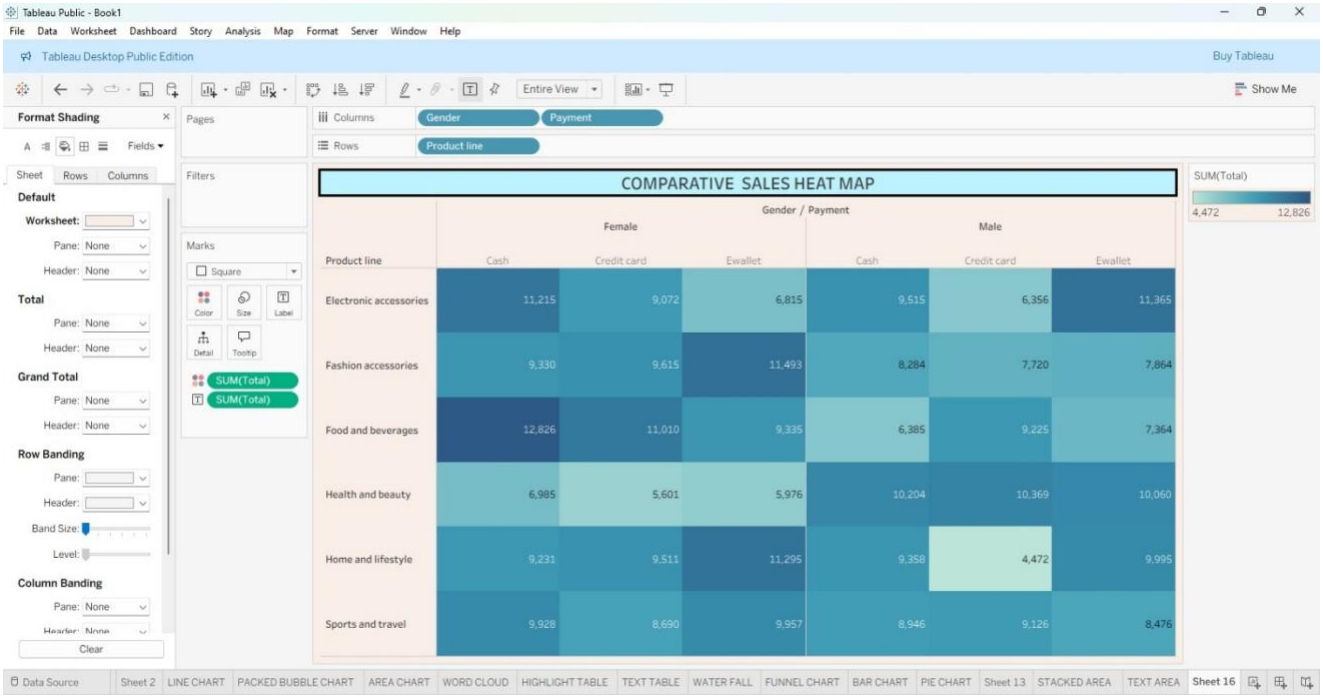
Step 2: Drag Product line →Rows

Step 3: Drag Branch → Columns

Step 4: Drag Total → Marks → Colour on the Marks card, Click the drop-down → Select Square

Step5: In Marks card → Size Drag slider to the right until squares fill the cells. Step 6:Click Colour →Edit Colours Choose a colour palette -Click Apply →Ok

Highlighted table: comparative sales Heat Map



5.PrepareWordCloud:

Step 1: Create a New Sheet and Renamed as “Word Cloud”

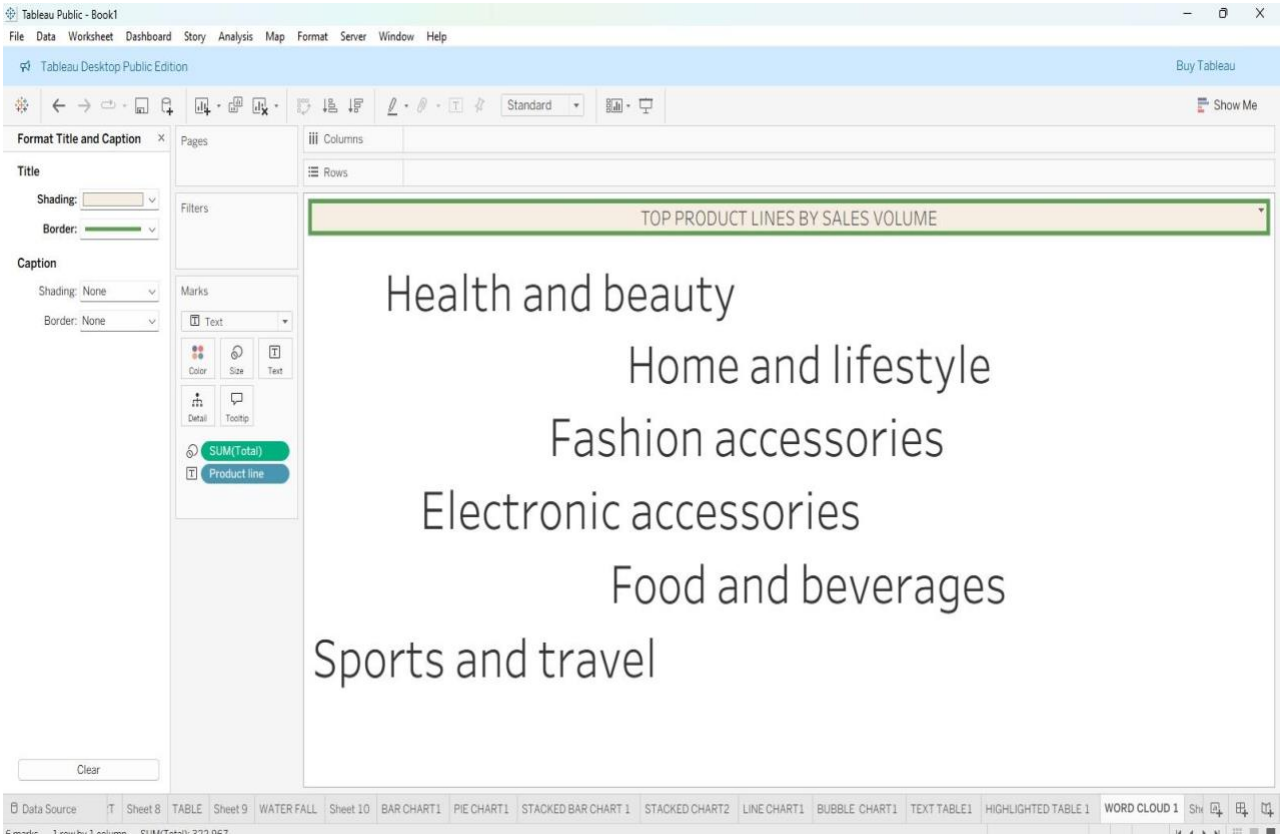
Step 2: In the Marks card -Click the drop down-Select Text.

Step 3: Drag Product line → Drop on Text Click Text → Increase font size

Step 4: Drag Tota l→ Drop on Size -Click Size → Increase slider

Step 5: Drag Total → Drop on Colour

Word cloud: Top Product Lines By Sales Volume



6.prepare Funnel Chart:

Step–01: Create a New Sheet and Renamed as “Funnel Chart”

Step–02: On Marks Card-Select Bar

Step – 03: Drag Measure Values → Columns-Edit filters

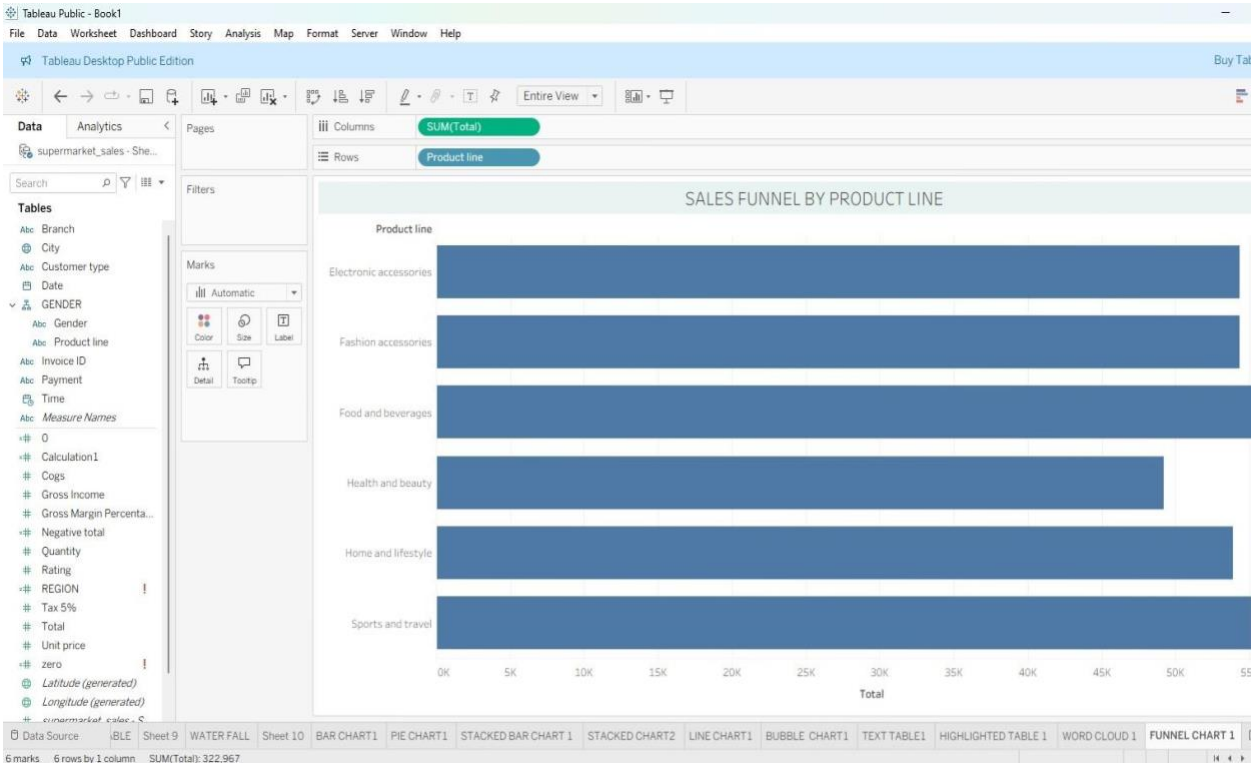
Step – 04: Drag Measure Names→ Rows-It automatically shows in filter

Click Sort →Descending (largest to smallest)

Step–05: Right click on sheet-click on Format-Go to Format Shading-click on

Worksheet -change colour.

Funnel Chart: Sales Funnel By Product Line



7.prepare Waterfall Chart:

Step 1: Create a New Sheet and Renamed as “Water fall Chart”

Step 2: Drag Product line →Columns-Select All

Step 3: Drag Running Total→ Rows

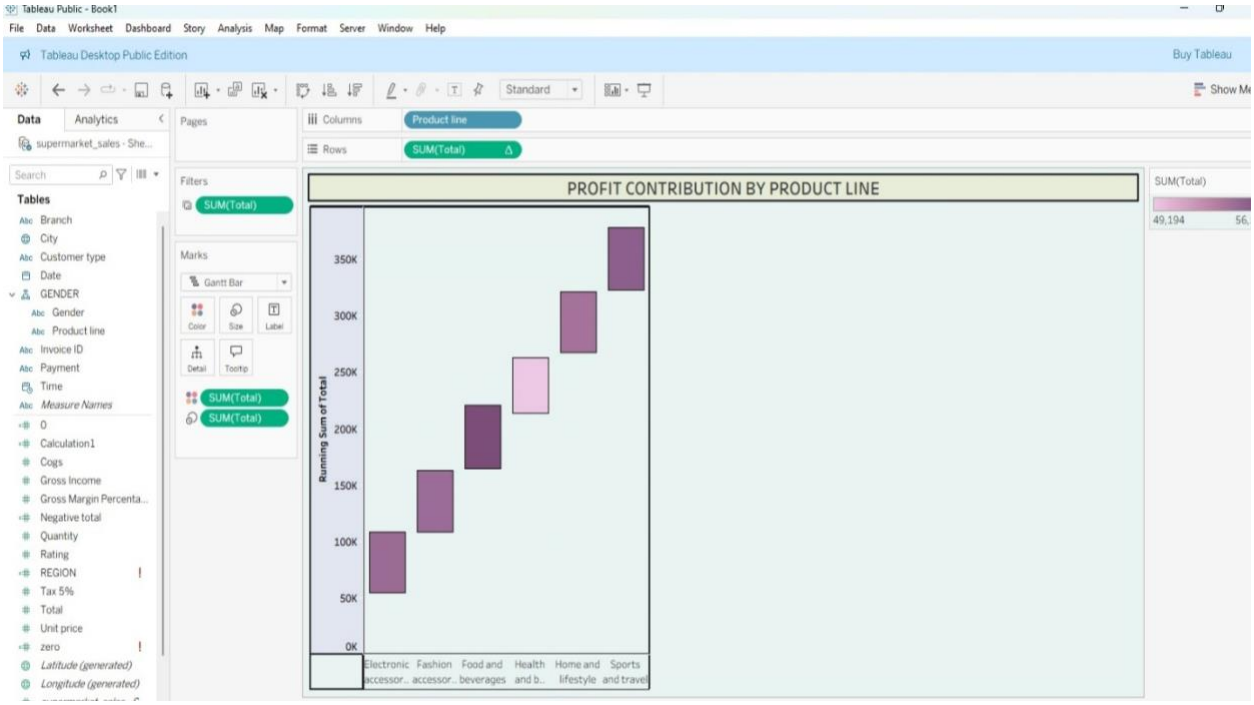
Step 4: Click Show Me-Select Gantt Bar

Stepn5: Right-click →Create Calculated Field Name: Difference

Formula: SUM([Total])-Click OK

Step 6: Drag Difference → Size (Marks card) Step–07:

Drag Difference →Colour

Waterfall Chart

Waterfall: Branch wise contribution to gross income